

How Do Price Surcharges Impact Water Utility Financial Incentives to Pursue Alternative Supplies during Drought?

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Abstract: Municipal water utilities commonly manage drought by encouraging conservation. Efforts to temporarily reduce water consumption can cost significantly less than investments in infrastructure to increase supply. However, reduced water sales during an extended drought can hinder a utility's ability to recover its (mostly fixed) costs, potentially leading to lower credit ratings and higher bond financing rates. Drought surcharges have emerged as an increasingly common solution to this problem. Compared to typical restriction-based policies (i.e., residential irrigation mandates), pricing incentives increase the economic efficiency of conservation while also mitigating some or all lost utility revenue. However, drought surcharges can also provide utilities with financial disincentives for pursuing cost-effective alternatives to conservation, like water transfers or wholesale (treated water) purchase agreements, when they are available. Wholesale water purchases between neighboring water systems are playing a more prominent role within urban water management plans, particularly in growing metropolitan regions of the southeastern United States. By reducing the need for conservation during drought, wholesale purchases can help utilities avoid welfare losses associated with consumption reductions. Using simulations of changing consumption patterns obtained from the detailed billing data of a mid-sized water utility, we found that the use of surcharges can cause conservation to appear more financially attractive to utilities than wholesale purchase agreements, even when the consumer welfare losses from conservation are larger than purchase costs. Results illustrate how the pursuit of efficiency with respect to one policy option (i.e., conservation) can alter the incentives to integrate that policy with a wider range of potentially beneficial alternatives. DOI: [10.1061/\(ASCE\)WR.1943-5452.0001228](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001228). © 2020 American Society of Civil Engineers.

Introduction

In the relatively water-rich eastern United States, utilities have aimed to maintain large volumes of infrequently used supply capacity as insurance against drought. Increasing development costs and onerous regulatory approval processes associated with supply expansion (Scudder 2005) have led to the increased use of temporary, adaptive alternatives for managing scarcity under continued population growth (Gleick 2003; Kundzewicz et al. 2007). Traditionally, utilities have used conservation strategies (e.g., outdoor use restrictions) as their primary response to temporary water shortages, but in many cases there are also opportunities to augment supply via water transfers (Vaux and Howitt 1984; Lund and Israel 1995; Wilchfort and Lund 1997; Green and Hamilton 2000; Jenkins and Lund 2000; Jaber and Mohsen 2001; Calatrava and Garrido 2005; Zarghami et al. 2010) or wholesale purchase agreements with other users (Isley and Middleton 2003; Reedy and Mumm 2012; Asefa

et al. 2014; Tran et al. 2019). Temporary policies can also be financially destabilizing because they lead to potentially large and intermittent revenue reductions or cost increases that are difficult to account for in annual budgets (Zeff and Characklis 2013; Leurig 2012; Hughes and Leurig 2013). In contrast, expensive but financially stable infrastructure-centric strategies are dominated by debt service payments.

Many water utilities' budget models revolve around cost recovery, setting water rates at levels expected to return revenues roughly equal to their (largely fixed) costs, which are often dominated by debt service on infrastructure (Tiger 2000; Bishop and Weber 1996). When determining a pricing structure, water utilities often employ a so-called two-part tariff (Brown et al. 1992; Griffin 2001), composed of a fixed service fee and a volumetric fee that is based on total consumption at the tap. Rate structures that generate a higher proportion of revenue through fixed fees can provide revenue stability even with unpredictable consumption, but the low volumetric charges send a weak conservation message to consumers (Mohayidin et al. 2009). Recent trends in water utility pricing have seen a larger proportion of revenues generated through volumetric charges (i.e., increasing block tariffs) (Tucker et al. 2015), making revenues more sensitive to consumption. Less conventional pricing structures like water-budget-based block pricing have attempted to address this revenue variability while also maintaining price incentives for conservation (Baerenklau et al. 2014), but more complicated water budget structures can be costly to implement and harder for consumers to understand. Even in water-scarce California, over 50% of water utilities had implemented some type of block pricing structure as of 2008, but only a small percentage of those had incorporated household-level water budget measurements into pricing (Mayer et al. 2008) for reasons that we do not explicitly examine here. This can create a risk of budgetary shortfalls for water utilities because their predominately fixed costs are independent of consumption. Long-term trends in revenue or cost

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are typically mitigated through changes to water rates, but the lengthy political and/or regulatory processes (Priest 1993) are not well suited to bridge short-term budgetary gaps. The resulting budget variability can negatively affect utility credit ratings (Scott et al. 2011), increasing debt service payments, which can make up a significant portion of overall costs in the capital-intensive water sector. These issues are becoming particularly worrisome for utilities as credit ratings agencies have begun to take interest in the hydrologic causes of revenue variability (Standard & Poor's 2012; Chapman et al. 2016). If, as seems likely, utilities make use of short-term responses more frequently in the future (Postel 2000; Gleick and Palaniappan 2010; Vorosmarty et al. 2000; Falkenmark 1999), the prospect of a credit rating downgrade and higher interest rates could give utilities an incentive to design their policies in a way that limits revenue and/or cost variability.

Conservation measures also impose losses on consumers who are expected to use less water during drought. Utilities address consumer surplus losses during periods of temporary conservation by (1) targeting "low-value" water uses with mandated restrictions (i.e., outdoor water use regulations), (2) incentivizing consumption reductions with preapproved drought surcharges that are implemented under conditions of scarcity, or (3) some combination of the two. Drought surcharges provide a more economically efficient means of enacting conservation than mandated usage restrictions, resulting in smaller welfare losses from conservation than mandated water use restrictions alone (Sibly 2006; Grafton and Ward 2008; Krause et al. 2003; Woo 1994; Smith and Wang 2008). In addition, the price inelasticity of urban water demands means that surcharges often provide utilities with additional revenue to compensate for a lower volume of sales, reducing potential budgetary disruptions. When there are no options to purchase temporary supplies during drought, replacing mandated restrictions with drought surcharges designed to achieve the same level of consumption reductions will improve measures of both consumer surplus and budgetary stability, provided that any additional revenue received by the utility is returned to consumers in a way that is reflected in the measurement of consumer surplus. However, the ability to make wholesale water purchases introduces a potential tradeoff between measures of budgetary stability and overall welfare (i.e., delaying the imposition of drought surcharges with wholesale purchases can increase welfare while simultaneously increasing budgetary instability). This case study demonstrates how drought plans that prioritize surcharge-induced conservation over wholesale purchase agreements may be allowing budget stability objectives to assume precedence over aggregate measures of welfare.

This analysis uses pricing and water use data from the Orange Water and Sewer Authority (OWASA) in North Carolina to investigate how drought surcharges affect utility revenue, consumer surplus related to water consumption, and utility incentives to pursue wholesale purchases. Using 5 years of individual monthly billing records, we develop a model of how drought surcharges interact with mandated restrictions at the account level and the resulting revenue implications. Results illustrate how the pursuit of improved efficiency with respect to one policy option (i.e., water conservation policies) can skew the incentives to pursue an effective mix of policy alternatives in a regulated environment.

Portfolio-Based Drought Management Policies and Institutional Inertia

Recent studies in the field of integrated water resources management have demonstrated how drought management policies that employ diversified "portfolios" of water management options can

provide greater adaptability to highly uncertain future conditions with a smaller infrastructure footprint and lower overall costs (Characklis et al. 2006; Kirsch et al. 2009; Kasprzyk et al. 2009; Kidson et al. 2013; Matrosov et al. 2013, 2015). In particular, studies focused on the Research Triangle region of North Carolina (including OWASA and its service area) found that integrating wholesale water purchases from a regional source (Jordan Lake) improves a wide range of short-term drought objectives, including reliability, expected cost, financial risk, and measures of the burden of conservation placed on consumers (Zeff et al. 2014; Herman et al. 2014). In the longer term, regional infrastructure to increase systemwide conveyance capacity reduces investment costs and provides managers with more flexibility to address uncertainties in climate and demand growth (Herman et al. 2015; Zeff et al. 2016).

Current drought management policies for Research Triangle water utilities under utilize even the current system capacities in the region (OWASA 2010; Goodwin and Cefalo 2010; Westbrook et al. 2009; City of Raleigh 2010). In particular, Jordan Lake contains municipal water supply allocations for many local utilities (including OWASA), but only one water treatment plant, operated by the Town of Cary, has direct access to the resource. During drought, water can be conveyed through interconnections between treated water distribution systems to augment existing, independent water supply systems for each utility. In addition to the volumetric costs of treating and pumping the purchased water, wholesale rates reimburse Cary for rented capacity in their treatment/conveyance infrastructure. Although Cary's system is designed to have excess capacity available for rent under typical conditions, OWASA prioritizes conservation measures over the use of this treatment/conveyance infrastructure in its published drought management plans.

The actions of OWASA during 2002–2008 highlight the adverse implications of these policies on consumers. The OWASA system serves 80,000 customers in Chapel Hill and Carrboro, North Carolina; OWASA can implement conservation via direct restrictions on use and drought surcharges and can enter into temporary wholesale purchase agreements with its neighbors. After the construction of a new reservoir in 1990, OWASA enjoyed relative water abundance, and neither conservation nor wholesale purchases were needed during 1990–2002 after imposing water use restrictions six separate times during the 1980s. In 2002, the combination of population growth and an extreme drought caused the utility to enact water use restrictions during 4 months in the late summer/early fall. The reduced water consumption observed during the drought initiated budgetary shortfalls that led to a credit rating downgrade in 2006. Instead of pursuing agreements with neighboring utilities to secure wholesale purchases from nearby (and underutilized) Jordan Lake, OWASA implemented a schedule of drought surcharges. Although these surcharges are estimated to reduce surplus losses compared with conservation achieved through mandatory restrictions alone, they also remove most of the financial benefit the utility generates by offsetting conservation with wholesale purchases. When another drought occurred in 2007–2008, OWASA spent 8 months under conservation, but did not purchase any water, despite availability through interconnections with neighboring utilities. The revenue losses from conservation were minimal, and the ability to weather a drought while maintaining revenue stability likely played a role in OWASA's credit rating upgrade 2 years later. However, consumers were forced to bear the burden of 8 months of conservation. Even now, OWASA's water shortage response plan dictates that purchase agreements will not be executed until the utility is already under conservation, even though they are still available for purchase at a price far less than OWASA charges its retail customers.

Welfare and Revenue Effects of Rationing by Price and Fiat: A Conceptual Framework

To understand the effect of drought surcharges on the incentives of utilities, we first consider a simplified example where water use is divided into two homogenous classes, indoor and outdoor (Fig. 1), subject to the same constant volumetric water price, P_1 . The price P_1 is set equal to the average cost of water provision (plus a regulated rate of return), a situation consistent with standard practice (Brown et al. 1992; Griffin 2001). In fiscal year 2018, the marginal cost of water delivered to consumers (chemicals and utilities for water treatment and distribution) accounted for only 5.75% of OWASA's total operating costs (OWASA 2018), a typical fraction for water utility budgets dominated by fixed costs (Tiger 2000). Because the reduced use of power and chemicals is likely to have a small and consistent effect under all drought management policies (reducing consumption via mandated restrictions or drought surcharges, and purchasing treated pumped water from Cary's water treatment plant), it is assumed that the marginal cost of treated water production at all relevant consumption levels is zero for illustrative simplicity. Different price elasticities of demand are used to represent typical differences between the two usage classes (higher elasticity for outdoor uses). In Fig. 1(a), mandatory water use restrictions of the amount ΔQ_{O1} are applied to the outdoor use class (assumed to be allocated efficiently across the outdoor uses), leading to a quantity of lost utility revenue and lost consumer surplus. The quantitative measures illustrated in Fig. 1 approximate welfare changes as areas under and behind demand curves for water. The welfare justification for consumer surplus is well known (Willig 1976), and for goods that comprise small budget shares for consumers and/or have small income elasticities of demand, changes in consumer surplus are close approximations to the more rigorously interpretable willingness to pay and willingness to accept measures. Both budget shares and income elasticities for water are small (Dalhuisen et al. 2003), making any approximation error

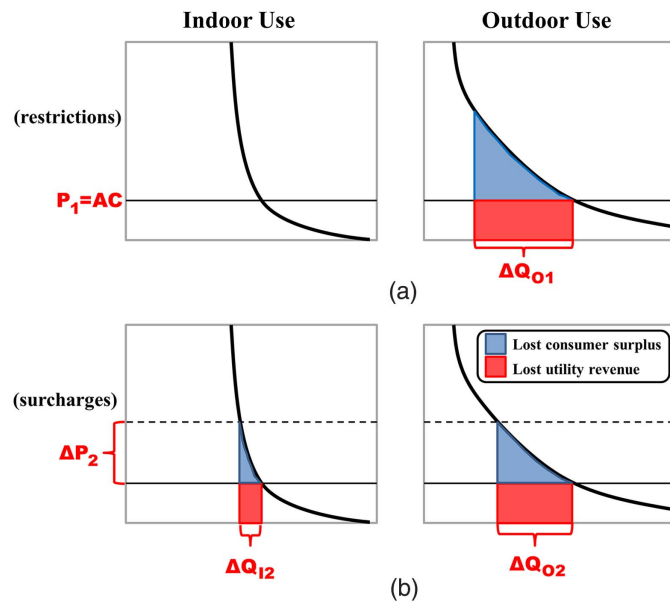


Fig. 1. Lost utility revenue and consumer surplus resulting from (a) reductions in water consumption from mandatory restrictions on outdoor use (Q_{O1}); and (b) water consumption reductions caused by drought surcharges to indoor (Q_{I2}) and outdoor (Q_{O2}) use. Price is assumed to be constant at all consumption levels, equal to the average cost of water provision.

particularly small in this case. Areas under appropriately defined inverse demand curves have the same welfare justification (Madden 1991).

The homogeneity of outdoor water users is a key assumption in Fig. 1, one that suggests that mandatory outdoor usage restrictions will limit only the least-valued outdoor uses, mostly related to lawn irrigation. This simplification underestimates the actual consumer surplus losses from mandatory restrictions to the extent that outdoor uses are not entirely interchangeable. For example, individual consumers may value car washing more than lawn watering or using a swimming pool. A more realistic conceptual model is used in our empirical application, which considers water demand for a larger variety of water uses.

As an alternative to restrictions, conservation can also be achieved through the application of a drought surcharge [Fig. 1(b)], ΔP_2 , resulting in a new water price $P + \Delta P_2$ for both uses, chosen so that reductions to both indoor (ΔQ_{I2}) and outdoor uses (ΔQ_{O2}) sum to equal the utility target consumption reductions caused by mandatory restrictions (ΔQ_{O1}). Reductions achieved through surcharges result in surplus losses in both indoor and outdoor uses, as well as utility revenue losses. As one would expect, the loss in surplus from drought surcharges summed across both user classes is smaller than the lost surplus observed in the case of mandated restrictions [Fig. 1(a)], as the same volume of water use reductions is more efficiently distributed across the two classes through pricing signals.

Assuming inelastic municipal demand surcharges will generate additional utility revenue [Fig. 2(a)]. This revenue is equivalent to the reduction in consumer surplus from the remaining water consumption, effectively resulting in the transfer of some consumer surplus to utility revenues.

Utilities often use a combination of drought surcharges and mandated restrictions [Fig. 2(b)] to achieve the targeted level of conservation ($\Delta Q_{I3} + \Delta Q_{O3}$) as a means of limiting the size of the surcharge. Mandated restrictions reduce outdoor use by a quantity ΔQ_{O3} , at which point the new price $P + \Delta P_3$ is applied to both indoor and outdoor use. The price increase causes a reduction

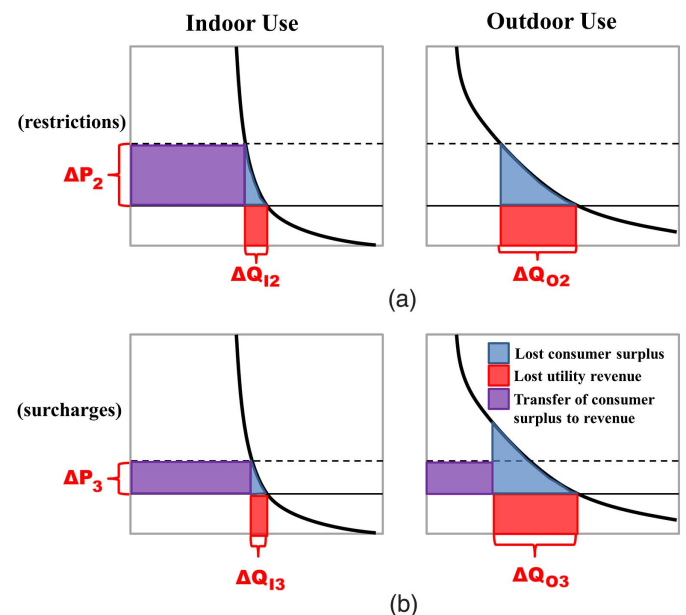


Fig. 2. Transfer of consumer surplus to utility revenue when consumption reductions are caused by (a) surcharges only; and (b) surcharges and mandatory use restrictions.

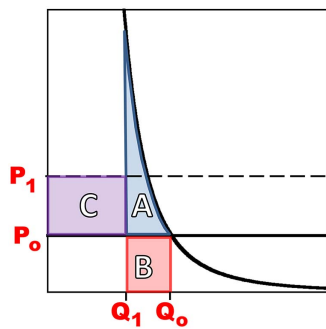


Fig. 3. Lost consumer surplus from foregone use. Area A indicates lost utility revenue; area B indicates transfer of consumer surplus to utility revenue; and area C indicates when surcharges and mandatory restrictions are used to achieve a given quantity of consumption reduction for a single use type.

in indoor water use by a quantity ΔQ_{I3} . Assuming hypothetical restrictions on outdoor use that successfully target the lowest consumer-value outdoor uses, the marginal value of outdoor water consumption at the new, restricted levels is higher than the new price ($P + \Delta P_3$), causing surcharges to have no further effect on outdoor use.

This two-instrument policy addresses a measurement and transaction cost problem, allowing utilities to use restrictions to control lower-valued, easily enforced outdoor use limits and to use drought surcharges to reduce the less observable indoor use of water. When applied to a diverse customer base, a fixed reduction in water consumption ($Q_0 - Q_1$) will result in some magnitude of lost consumer surplus, lost utility revenue from reduced sales, and transfer of consumer surplus to utility revenue (Fig. 3). Policies that favor mandated restrictions will impose larger consumer surplus on high-volume users with substantial outdoor water use that is easily targeted. The introduction of surcharges shifts some of this burden away from high-volume users (specifically, users who assign higher than normal value to their outdoor water uses) and toward lower-volume users, such as those living in multifamily units. Low-volume users observe losses to consumer surplus in two ways: (1) by foregoing uses that they deem lower value than the new surcharge prices, and (2) by paying higher prices for high-value uses of water (e.g., cooking, cleaning, drinking) that continue to be exercised during drought. The proportions of overall conservation attributed to surcharges and mandated restrictions have an effect on the relative impact of each effect. The previous literature (Sibly 2006; Grafton and Ward 2008; Krause et al. 2003; Woo 1994) focused on the ability of drought surcharges to reduce the surplus or total welfare losses, represented as the sums of reductions to consumer surplus and reductions in revenue from lower water consumption. Here, we additionally consider how surcharges affect utility revenue, represented as the difference between revenue reductions from lower water consumption and revenue increases from higher prices. For a given reduction in water consumption (ΔQ), increasing the fraction of reduction attributable to surcharges (and subsequently reducing the mandated restrictions) will increase the size of revenue increases from higher prices, reducing utility revenue losses, and potentially even turning them into revenue gains.

Empirical Measurement

OWASA (2010) outlines how conservation targets are expected to increase as drought conditions worsen. These targets are organized

into stages, which are triggered using reservoir storage thresholds that vary throughout the year based on changing hydrologic and demand patterns. The revenue and consumer surplus losses from Stages I and II conservation targets are calculated using individual water consumption data from 2002 through 2006. Stages I and II were chosen here because they correspond to the levels of restrictions implemented in both the 2002 and 2007/2008 droughts experienced by OWASA. Conservation targets are larger when outdoor water use is more prevalent, so simulations apply reduction targets from each of Stages I and II to a typical month in both the irrigation (July) and nonirrigation (January) seasons. OWASA expects to meet these targets through a combination of mandated restrictions and drought surcharges.

Wholesale water purchases are available to OWASA through interconnections with another municipal water provider in the region, the Town of Cary. An existing agreement between the two utilities would provide OWASA with treated drinking water at a cost of \$0.79/m³. The water source for these agreements, Jordan Lake, is operated by the USACE to ensure that flood control guidelines and environmental requirements are met on the Cape Fear River. USACE has also designated a portion of the storage in Jordan Lake for municipal water supply, providing multiple local utilities (including OWASA) with individual allocations to the storage pool. However, the Town of Cary operates the only raw water intake and water treatment plant on the lake. Cary is able to withdraw water allocated to OWASA, treat it, and then pump it through direct interconnections into OWASA's distribution system. Wholesale purchase costs do not reflect the value of the "raw" water; instead, they essentially serve as a fee to cover treatment and pumping costs as well as the rental of capacity within the Cary water treatment plant. The OWASA allocation in Jordan Lake is equivalent to about 18,939 m³ of water per day, or 60% of OWASA's average daily demand. During the summer months, when drought policy is typically implemented, OWASA's demands are higher, but this interconnection still represents significantly more water than the expected conservation from its Stage II restrictions (30% of overall demand).

We implemented a welfare measurement model using detailed billing data from OWASA. Using monthly account billing data, we modeled the response of individual OWASA customers to the imposition of surcharges and water use restrictions. Contrasting with the simple example from the section describing our conceptual framework, the individual billing data used in this analysis divides all OWASA customers into 21 groups, described in Table 1. The groups are distinguished by an elasticity value and a binary attribute indicating whether or not mandatory restrictions apply to the group. For each bill, we calculated individual reductions in consumer surplus ($A + C$ in Fig. 3) and utility revenue ($B - C$ in Fig. 3) during periods of conservation. Because the marginal cost of treated water production is assumed to be zero, an approximation to welfare change should consider the sum of these two losses, the entire area under the demand between the original (Q_0) and modified (Q_1) water consumption levels (i.e., the area $A + B$ in Fig. 3). To determine individual consumer response to price changes, price elasticities were taken from an econometric study by Burton and Associates (2007). The study used regression analysis on customers with uninterrupted billing records from 2001 to 2005 to develop a relationship between water consumption and price, controlling for a number of weather variables and local unemployment (as a stand-in for economic conditions). The study specified 18 unique water demand models to group consumption by residential and nonresidential classes, high, medium, and low water use, and three distinct meter-reading cycles. In the study, elasticity values from each water demand model (Table 1) are matched with account billing data used

Table 1. Water user groups and attributes

Demand model	Use range (m ³)	Elasticity	Restrictions
Low use res., meter cycle 1	<15.2	−0.03	No
Low use res., meter cycle 2	<15.2	0.0	No
Low use res., meter cycle 3	<15.2	−0.19	No
Medium use res., meter cycle 1	15.2 < x < 22.7	−0.62	Yes
Medium use res., meter cycle 2	15.2 < x < 22.7	−0.50	Yes
Medium use res., meter cycle 3	15.2 < x < 22.7	−0.61	Yes
High use res., meter cycle 1	>22.7	−1.15	Yes
High use res., meter cycle 2	>22.7	−1.09	Yes
High use res., meter cycle 3	>22.7	−1.24	Yes
Multifamily, meter cycle 1	All	−0.03	No
Multifamily, meter cycle 2	All	0.0	No
Multifamily, meter cycle 3	All	−0.19	No
Low use comm., meter cycle 1	<45.5	0.0	No
Low use comm., meter cycle 2	<45.5	0.0	No
Low use comm., meter cycle 3	<45.5	0.0	No
Medium use comm., meter cycle 1	45.5 < x < 182	−0.36	Yes
Medium use comm., meter cycle 2	45.5 < x < 182	−0.33	Yes
Medium use comm., meter cycle 3	45.5 < x < 182	−0.60	Yes
High use comm., meter cycle 1	>182	−0.37	Yes
High use comm., meter cycle 2	>182	−0.61	Yes
High use comm., meter cycle 3	>182	−0.38	Yes

Note: res. = residential; and comm. = commercial.

to determine individual consumer response to increasing drought surcharges. The account billing data specify monthly consumption and a user-type that can be used to assign each bill to one of the 18 demand models specified in Table 1. The Water Shortage Response Plan of OWASA (2010) states that mandatory restrictions apply to all large commercial and industrial users of water as well as all irrigation and outdoor residential uses. For the purposes of this study, we consider any commercial or industrial use of over 45.5 m³/month to be subject to water use restrictions. Although OWASA does not differentiate between indoor and outdoor residential use for billing purposes, seasonal use patterns in the region support the assumption that the average residential user uses about 18.9 m³ per month on nonobservable indoor uses. For the purposes of this study, residential restrictions are modeled as reductions on any water consumption that exceeds 18.9 m³ per month on residential bills. It is assumed that users billed as multifamily residential users (i.e., apartments, dormitories) have minimal outdoor demands and are not subject to mandatory use restrictions, regardless of their total monthly consumption. During the study period, 2002–2006, OWASA charged its customers a seasonal pricing schedule, with higher rates during the summer irrigation months. This period was bracketed by two major droughts during which OWASA implemented periods of temporary conservation, and during this time OWASA developed the drought surcharge schedule that was employed during the latter drought. After 2007, OWASA also implemented a block pricing schedule, complicating measurement of surplus effects (Hewitt and Hanemann 1995). In addition to price levels, demand also changes over the course of the year, particularly for irrigation-heavy customers, and reductions from surcharges and water use restrictions are calculated in both January and July. Constant elasticity demand functions are constructed for each consumer group, detailed in Table 1

$$w_{i,h} = k_{i,h} * P_i^{e_g} \quad (1)$$

where i = month of the year; h = individual account index; g = consumer group; $k_{i,h}$ = constant calculated from account water

use data and price history; P_i = volumetric water price; $w_{i,h}$ = account water consumption (MG); and e_g = consumer price elasticity of demand in group g .

The constant $k_{i,h}$ in Eq. (1) is calculated using observed water consumption from individual billing records and historical records of price, with estimates of elasticity based on the consumer groups outlined in Burton and Associates (2007). Once $k_{i,h}$ is calculated, changes to individual account consumption based on price changes can be estimated. Changes to surplus ($A + C$ in Fig. 3) and utility revenue ($C - B$ in Fig. 3) are calculated for different combinations of two parameters: (1) overall reduction targets, and (2) the proportion of the overall reduction directly caused by surcharges. For example, if the overall reduction target is 15% of water consumption (a typical drought response level), and two-thirds of the reductions come from surcharges, then a surcharge is calculated that will result in a 10% reduction in overall water consumption. Restrictions on residential outdoor, commercial, and industrial use are then increased until the total reduction is equal to 15% of the original water consumption. Surcharges are applied as a constant multiple of the volumetric water price, so that a surcharge of 25% will result in the volumetric water price increasing by a factor of 1.25. The surcharge level that achieves the desired target is calculated through iteration, converging on the reduction targets by solving a modified version of Eq. (1) for a surcharge value b , such that

$$w_{i,h}^* = \frac{1}{i} \sum_i \sum_h k_{i,h} \times (b * P_{i,h})^{e_g} \quad (2)$$

where $w_{i,h}^*$ = water use after application of surcharges; and b = drought surcharge.

Mandatory use restrictions are then imposed on all applicable consumers as described in Table 1. Reductions are assumed to be uniform across all consumers who are subject to mandatory outdoor use restrictions. Reductions from mandatory restrictions are measured with respect to normal (unrestricted) use levels, not surcharge-adjusted uses as calculated in Eq. (2). This allows the analysis to measure the reductions that would occur under a given level of mandatory restrictions without the impact of drought surcharges. We assume that mandatory restrictions on outdoor use are the least-valued uses of water at each billing location. Therefore, the pricing incentive will cause outdoor water customers to forgo the same consumption types targeted by mandated restrictions. This assumption, necessary for the analysis of the given dataset, introduces a bias that likely understates the welfare losses associated with mandatory restrictions, making them seem more favorable compared with drought surcharges. However, this effect is mitigated by distinguishing indoor versus outdoor use at the household level. For example, the benefits of moving from mandatory restrictions to drought surcharges are captured for households that would prefer to take shorter showers (reduce indoor use) and irrigate their lawn every day (maintain outdoor use), but not a household that would prefer to completely trade lawn irrigation (reduce outdoor use) for the ability to wash their car every day (maintain outdoor use).

Water consumption reductions caused by mandatory restrictions at an individual billing location are counted toward the overall consumption reductions only when they are greater than the reductions caused by surcharges at that same location (i.e., when the marginal value of water consumption at the restricted levels is higher than the price with surcharges, as in Fig. 2(b)).

The magnitude of mandatory use restrictions is calculated via iteration until the total consumption reductions across all consumers (surcharges plus mandatory restrictions) reaches the targeted level, such that

$$w_{i,h}^{**} = (1 - r) \times w_{i,h} \quad (3)$$

subject to

$$w_{i,h}^{***} = \begin{cases} w_{i,h}^* & \text{if } w_{i,h}^* < w_{i,h}^{**} \\ w_{i,h}^{**} & \text{if } w_{i,h}^* > w_{i,h}^{**} \end{cases} \quad (4)$$

where $w_{i,h}^{**}$ = water use after imposition of mandatory restrictions; $w_{i,h}^{***}$ = targeted overall water use during conservation period; and r = magnitude of mandatory restrictions.

Once the surcharge level b and the reduced total water consumption for each consumer $w_{i,h}^{***}$ are determined, lost utility revenue and consumer surplus can be determined. These values are calculated for each consumer and summed to arrive at a total value

$$LR_i = \sum_h w_{i,h} \times P_i - w_{i,h}^{***} \times P_i \times b \quad (5)$$

where LR_i = lost revenue from reduced water consumption; and P_i = volumetric water price.

Finally

$$LW_i = \sum_h \int_{w_{i,h}}^{w_{i,h}^{***}} \frac{w}{k_{i,h}} \frac{1}{\varepsilon_g} dw \quad (6)$$

where LW_i = lost welfare from reduced use.

Results and Discussion

Without the use of drought surcharges, OWASA revenue losses were estimated to be \$1.45/m³ of conservation during January and \$2.06/m³ during July, equal to the average price per m³ paid by consumers during 2002–2006 (Fig. 4). Estimates of total welfare

losses, including revenue losses and consumer surplus losses as in Eq. (6), ranged from \$1.81/m³ (January, Stage I) to \$8.54/m³ (July, Stage II). Surplus losses (per volume consumed) increase at higher conservation stages because more valuable uses of water (providing a larger consumer surplus) must be conserved to achieve the higher targets. Wholesale purchase costs (at \$0.79/m³) are equal to about half the revenue losses associated with mandated (no price incentive) conservation and as much as an order of magnitude less than the consumer surplus losses associated when cutbacks are significant (Stage II, July).

When surcharges are used to replace mandated restrictions, conservation generates lower lost revenues and consumer surplus. Revenue losses decrease steadily with the application of drought surcharges, and reach zero when surcharges rise to levels that could account for between 75% and 87% of the total conservation, depending on the season and restriction stage. If surcharges increase beyond this point, utilities gain revenue during periods of conservation. Per-unit revenue losses (\$/m³) fall below the cost of wholesale water purchases at even lower surcharge levels, reaching \$0.79/m³ when drought surcharges rise to levels that could account for 40%–55% of the total conservation. At this point, enacting conservation becomes more financially attractive than wholesale purchases. Overall welfare improvements are also generated by moving from mandated restrictions to conservation, resulting in welfare losses that are 16% smaller under surcharges (in \$/m³) during January (Stage I) restrictions (targeting 7% use reduction) and up to 70% smaller during July (Stage II) restrictions (targeting 30% use reduction). As noted earlier, welfare losses from mandatory restrictions are understated because it is unlikely that mandatory restrictions on outdoor use exclusively target the lowest-value uses for each consumer (e.g., some users may prefer to stop washing their cars rather than brown their lawns). In this way, surcharges may improve welfare outcomes more than the 16%–70% reported

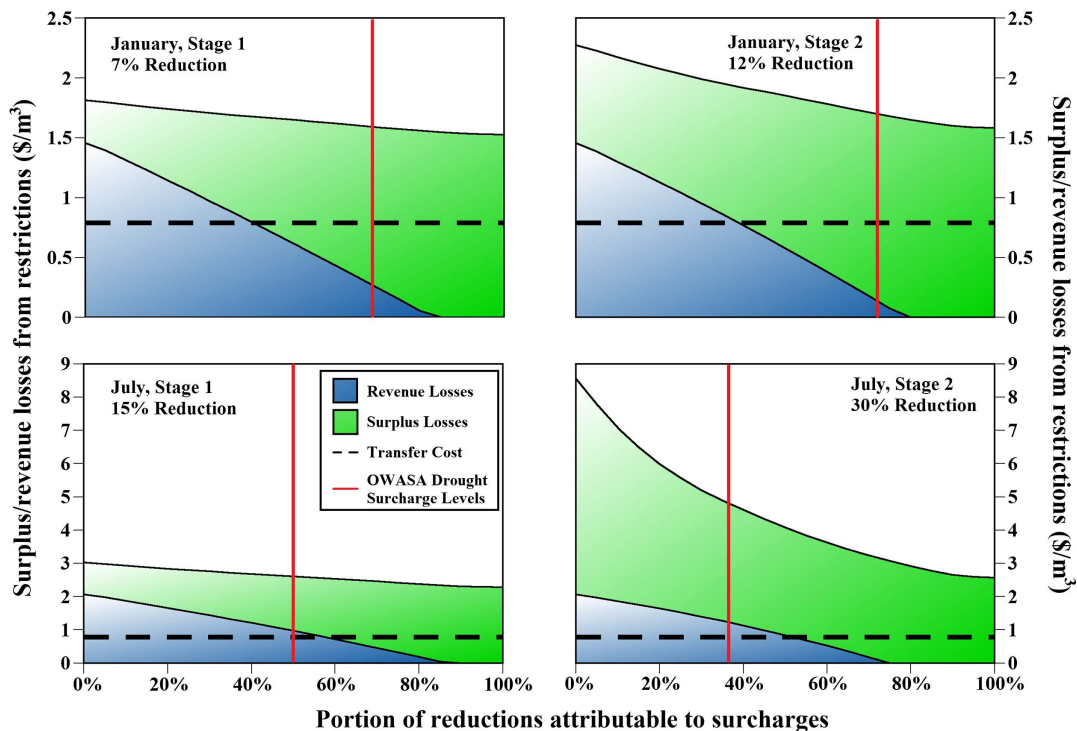


Fig. 4. Surplus losses and utility revenue losses (\$/m³) compared with transfer costs, as a function of the portion of reductions attributable to surcharges. Vertical lines show the level of water consumption reduction OWASA would expect to receive from surcharges alone. The four scenarios shown here represent the irrigation and nonirrigation season reduction targets for the first two stages of OWASA's drought management plans.

in this study. However, welfare losses from conservation remain 2–3 times as large as the cost of an equivalent volume of wholesale purchases, even when surcharges completely replace mandatory restrictions during drought. From a welfare perspective, wholesale purchases remain more beneficial than conservation during all seasons and stages evaluated here, even when it is achieved entirely through pricing incentives. Surcharges represent a win–win policy for the utility and consumers when conservation is the only viable response to drought, but not when alternatives to increase supply are available. Welfare losses from conservation remain much higher than the cost of purchasing the same volume of wholesale purchases under all conditions evaluated in this study, suggesting that consumers benefit from drought management plans that employ more purchase agreements. In the sense that surcharges discourage utilities from exercising purchase agreements in favor of implementing conservation, surcharges hurt the consumer, even though surcharges improve outcomes for the consumer during periods of conservation compared with mandatory restrictions on water consumption.

This effect also has the potential to retard investment in infrastructure that could be used to expand the capacity to enter into wholesale purchase agreements, such as the network of municipal interconnections OWASA uses to access larger regional water sources. The capacity of these pipelines constrains the rate at which neighboring supplies can be used as sources for purchase agreements, and upgrading them would entail significant costs. The benefits derived from these measures (avoiding welfare losses from conservation) could very well outweigh their upfront costs, but utilities must finance the upgrades with revenues generated from the sale of water. Water utility managers can raise rates to help pay for new infrastructure, but the ability to increase budgetary stability during droughts can also be a powerful incentive. The use of surcharges when conservation is enacted greatly reduces this incentive.

Drought surcharges can and do serve as a useful tool for municipal water utilities responding to a drought. However, this case study illustrates a scenario in which the budgetary stability provided by surcharges may lead a utility to undervalue wholesale purchase agreements with their neighbors, which can improve welfare but increase budgetary variability. In regions where greater interconnection between water systems could benefit consumers, a focus on maintaining financial stability during drought could lead utilities to inefficiently use regional water supplies. More explicit consideration of welfare effects during drought could also provide an incentive for greater investment in regional treatment and conveyance infrastructure that would further reduce the need for periodic conservation in the face of drought.

Conclusion

As the cost of new supplies continues to increase, temporary policies such as conservation or wholesale purchase agreements are playing a larger role in water utility efforts to manage drought, particularly in the eastern United States. The unpredictable and intermittent nature of the resulting financial implications (lost revenue, additional costs) can be difficult to manage, giving utility managers an incentive to develop plans that combine their use in a way that minimizes potential budgetary disruptions. Drought surcharges present utilities with a particularly compelling solution to this problem. In addition to mitigating some or all of these revenue losses, the pricing signals sent to consumers allow conservation to be distributed more efficiently than with simple restrictions on water use (e.g., residential irrigation mandates). However, drought surcharges

may also lead to unintended consequences for utilities focused on the budgetary impacts of conservation efforts, particularly when alternatives to conservation—such as wholesale purchases from neighboring utilities—are available.

Although conservation is viewed by utilities as a more conventional response to drought, particularly in the eastern United States, alternatives available at lower cost could provide an incentive to break with the status quo. Drought surcharges effectively make revenue losses from conservation “cheaper” than the cost of alternatives, even as the welfare losses from conservation remain significantly higher. In the case of OWASA, when surcharges account for 40%–55% of the total conservation, revenue losses fall below the cost of wholesale purchases from neighboring utilities (\$0.79/m³), even as welfare losses remain two to three times higher than these prices.

By prioritizing budgetary stability over estimates of welfare losses, utilities may bias themselves against alternatives to conservation, particularly when conservation is achieved using drought surcharges. Wholesale purchase agreements, if used at all, could be relegated to an “emergency” role, with conservation being used more frequently as a drought response. This would lead to unnecessary welfare losses from conservation in situations in which wholesale purchases could be available for a fraction of the resulting losses to consumers. With less of a financial incentive to avoid enacting conservation, surcharges could also retard investment in the infrastructure needed to access auxiliary supplies during drought, all to the detriment of consumers. These results illustrate how the pursuit of efficiency with respect to one policy option (i.e., conservation) can alter the incentives to integrate that policy with a wider range of potentially beneficial alternatives.

Data Availability Statement

Data, models, and code used in this study are part of a publicly available online repository that can be accessed at <https://github.com/hbz5000/TheRealTriangleWaterSupply>. Some data used in this study are proprietary or confidential in nature and may only be provided with restrictions.

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